Trace and Spectral Norms of Tensor Powers using Krylov Subspace Methods

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Introduction & Motivation

- The **Schatten** p**-norm** of $A \in \mathbb{C}^{N\times N}$ is the ℓ_p-norm of its singular values:

$$\|A\|_p = \left(\sum_{i=1}^{N} \sigma_i^p\right)^{1/p}$$

- $p = 1 \implies$ Trace Norm
- $p \to \infty \implies$ Spectral Norm (largest singular value)
- **The Challenge:** Computing exact norms requires Singular Value Decomposition (SVD), which costs $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ memory.

Target Operator & The Tensor Wall

- Our focus is on evaluating linear combinations of Kronecker powers:

$$M = c_1 A^{\otimes n} + c_2 B^{\otimes n}$$

- For 3×3 building blocks, the dimension grows exponentially: $N = 3^n$.
- The spectrum of such tensor networks is highly structured but rugged/clustered, causing standard decomposition solvers to stagnate or fail.
- Exact dense SVD becomes utterly uncomputable past $n = 9$ due to memory constraints.

Generalized Matrix Functions

- Instead of full SVD, we treat the norm estimation via *generalized matrix functions* ($f^\diamond(M) = Uf(\Sigma)V^\dagger$).
- The trace norm ($p = 1$) is equivalent to evaluating the trace of $\sqrt{M^\dagger M}$:

$$\|M\|_1 = \mathrm{tr}\left(\sqrt{M^\dagger M}\right)$$

- **Hutchinson's trace estimator** approximates this trace using s random vectors v_i:

$$\mathrm{tr}\left(\sqrt{M^\dagger M}\right) \approx \frac{1}{s}\sum_{i=1}^{s} v_i^\dagger \sqrt{M^\dagger M} v_i$$

Matrix-Free Krylov Projection & Quadrature

- Applying the Lanczos algorithm to an operator generates orthogonal polynomials with respect to its spectral measure.
- By projecting onto a low-dimensional Krylov subspace, we obtain a tridiagonal matrix T_ℓ.
- Evaluating the matrix function on T_ℓ is mathematically equivalent to an ℓ-point **Gaussian quadrature rule** over the spectral distribution.
- **Matrix-Free Advantage:** We only need an abstract subroutine for matrix-vector products $y \leftarrow Mx$. The massive $N \times N$ matrix is never explicitly stored!

Golub-Kahan Bidiagonalization (GKB)

- **Numerical Stability Issue:** Applying Lanczos directly to $M^\dagger M$ squares the condition number, leading to rapid loss of orthogonality.
- **Solution:** Project M directly using GKB, building bases U_ℓ and V_ℓ:

$$MV_\ell = U_\ell B_\ell, \quad M^\dagger U_\ell = V_\ell B_\ell^\dagger + \gamma_\ell u_{\ell+1} e_\ell^T$$

- The singular values of the small bidiagonal matrix B_ℓ serve exactly as the Gaussian quadrature nodes.
- **3-Term Recurrence:** GKB only needs the current, previous, and next vectors. The memory footprint remains strictly bounded at $\mathcal{O}(\mathbf{N})$.
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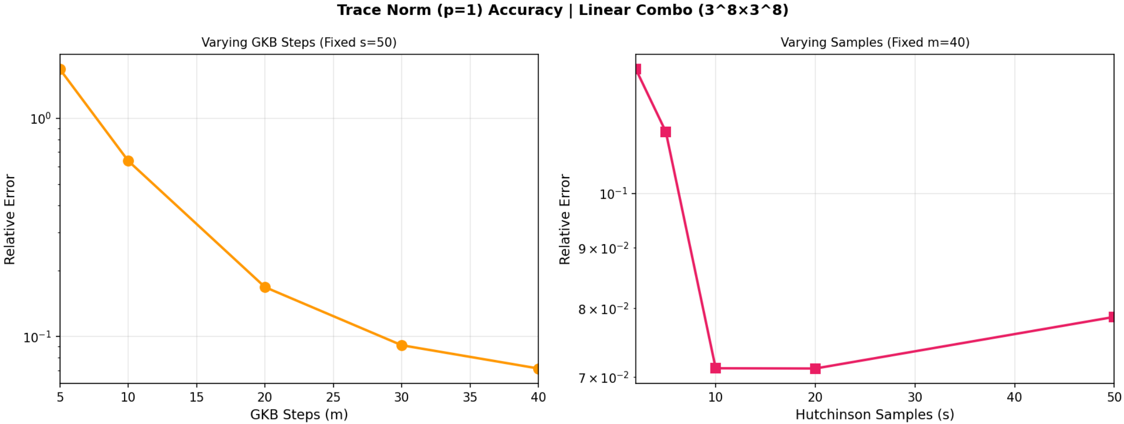
$$v^\dagger \sqrt{M^\dagger M} v \approx \|v\|_2^2 \left[\sqrt{B_\ell^\dagger B_\ell}\right]_{1,1}$$

Spectral Norm & IRLM

- For the spectral norm ($p = \infty$), we strictly seek the maximum singular value.
- Hutchinson stochastic averaging is completely unnecessary for this.
- We apply **Implicitly Restarted Lanczos Method (IRLM)** directly.
- The algorithm extracts **Ritz values** that dynamically approximate the extreme eigenvalues.
- IRLM converges exceptionally fast and is significantly more computationally efficient than a full GKB projection.

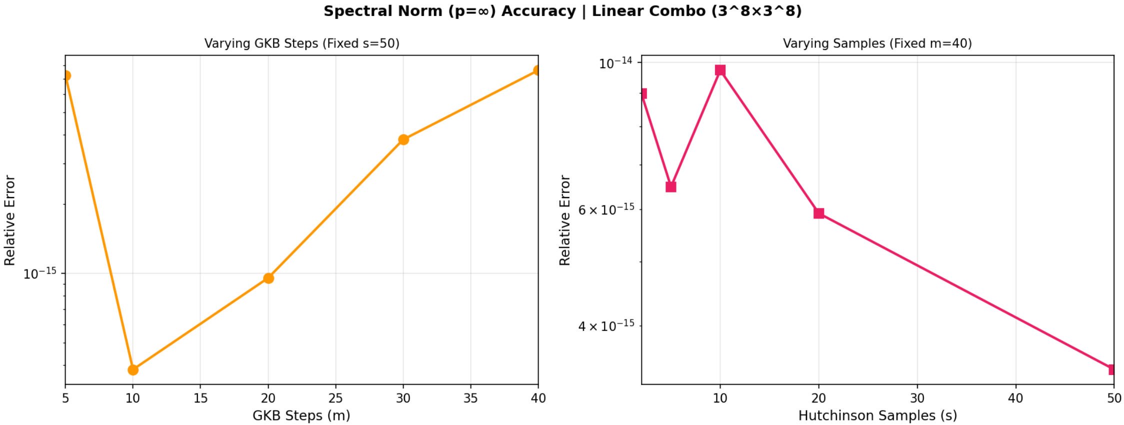
Accuracy Verification: Trace Norm ($n = 8$)

- Exact ground-truths acquired via SU(3) block-diagonal representations (`tensorpow`).
- At $n = 8$ ($N = 6,561$), convergence is sufficient using chosen s and m bounds.



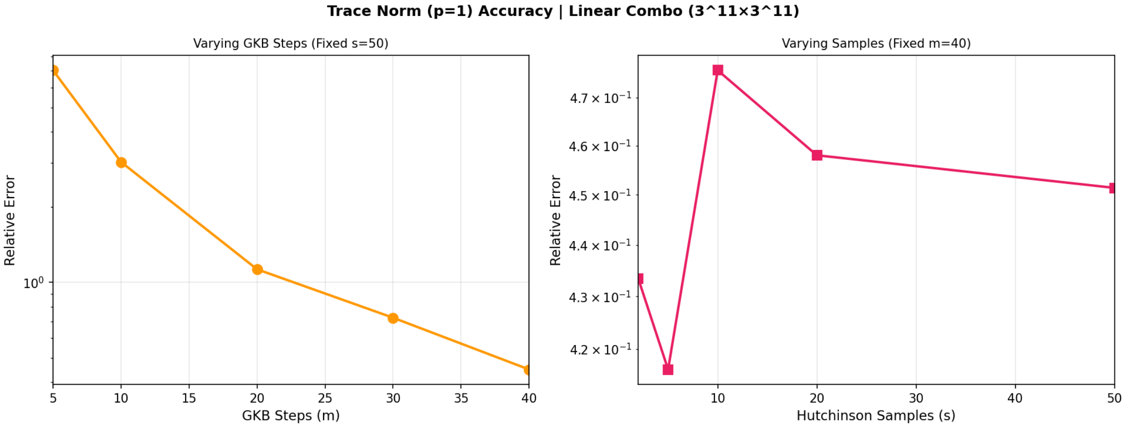
Accuracy Verification: Spectral Norm ($n = 8$)

- The spectral norm converges rapidly using the Implicitly Restarted Lanczos Method.



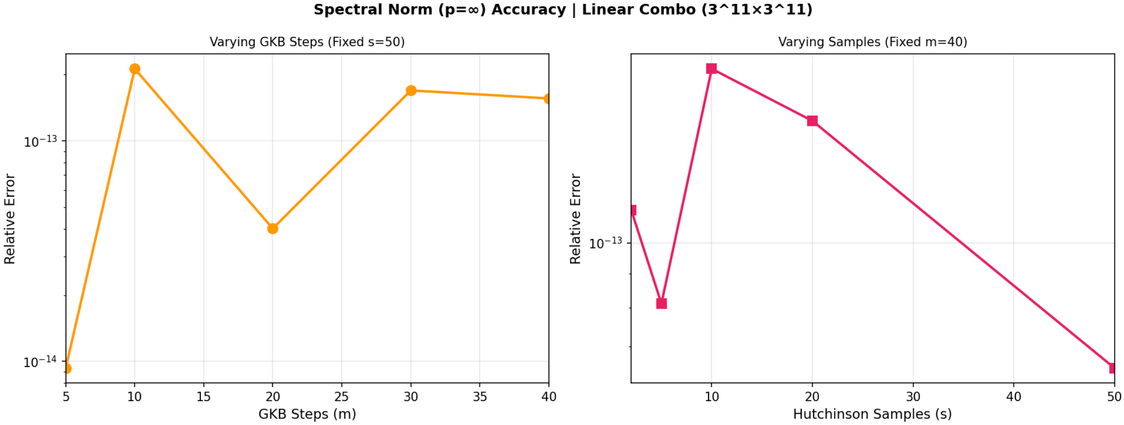
Accuracy Scaling: Trace Norm ($n = 11$)

- At $n = 11$ ($N = 177,147$), the error for $p = 1$ remains unacceptably high, indicating s and m must be increased.



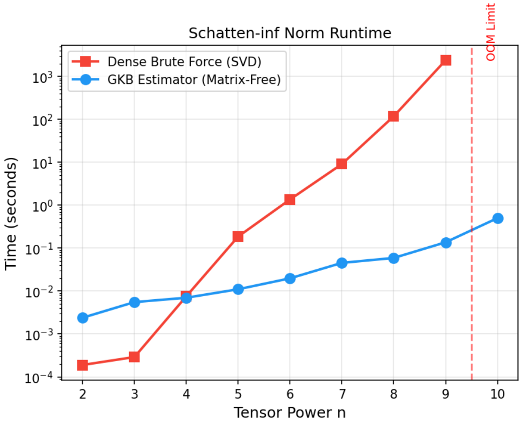
Accuracy Scaling: Spectral Norm ($n = 11$)

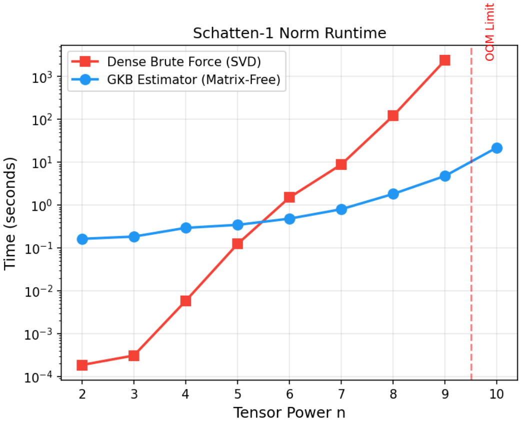
- The spectral norm continues to converge efficiently for larger tensor powers.



Performance Scaling Breakthrough: Computation Time

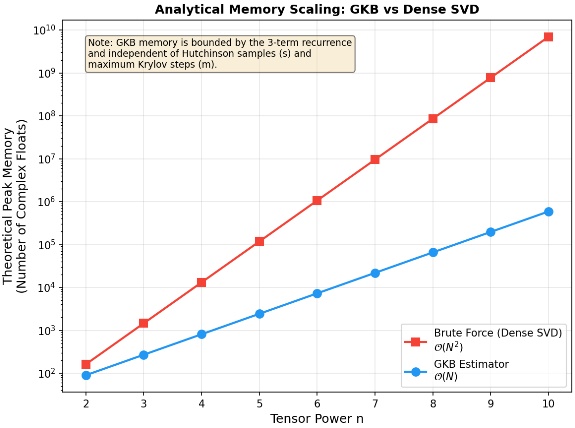
- Dense SVD takes 40 mins at $n = 9$. Krylov methods scale with $\mathcal{O}(N)$ instead of $\mathcal{O}(N^3)$.





Performance Scaling Breakthrough: Peak Memory

- Dense SVD triggers OOM at $n = 10$. The GKB matrix-free memory footprint remains bounded at $\mathcal{O}(N)$.



Possible Further Work

- Execute exhaustive large-scale experiments.
- Assess robust stopping criteria for arbitrary density matrices.

Thank You!